

NLP Assignment: 2

1. Explain the Concept of Regular Expression. Also explain basic operator (union, Concatenation, Star) and illustrate with examples how they generate regular language.

- Concept of Regular Expression
- i. A regular expression (RE) is a formal way to represent pattern in strings.
 - ii. It describes a set of strings called a regular language.
 - iii. Used in pattern matching, compiler and text processing.
 - iv. Defined over an alphabet.
 - v. Can represent infinite sets of strings.
 - vi. Built using symbols and operators.

* Basic Operators

I. Union

- i. Represent choice between symbols.
- ii. $a+b$ means either a or b
- iii. Language: $\{a, b\}$
- iv. Example: $(a+b) \rightarrow a \text{ or } b$
- v. Used for alternative
- vi. order does not matter
- vii. Generates union of languages.

* Concatenation

- i. Combines two expressions sequentially.
- ii. ab means a followed by b .
- iii. Language: $\{aby\}$
- iv. Example: $(a+b)c \rightarrow ac \text{ or } bc$

* Kleene Star (*)

- i. Represent repetition
- ii. a^* means zero or more a 's.
- iii. includes empty string ϵ .
- iv. language : $\{\epsilon, a, aa, aaa, \dots\}$
- v. Example: $(ab)^* \rightarrow \epsilon, ab, abab$

Q. Derive a Regular Expression for a given finite automaton. Explain the step-by-step process of converting a DFA/NFA into an equivalent regular expression with a suitable example.

Steps:

- i. Start with given DFA/NFA
- ii. Convert to Generalized Transition Graph (GTG)
- iii. Add single start and final state
- iv. Remove states one by one.
- v. update transition using RE rules
- vi. Continue until only start & final remain.
- vii. final edge give required RE

Example: DFA

- States : $q_0 \rightarrow q_1$
- Transition : $q_0 \xrightarrow{a} q_1$
- Loop : $q_1 \xrightarrow{b} q_1$

Step process:

1. Start at q_0 , go to q_1

2. with 'a'.
3. From 'a' loop with 'b*'.
3. Combine : ab^*

Final RE : ab^*

3. Construct a Regular Expression for the language $L = \{w \in \{a, b\}^+ \mid w \text{ ends with 'abb'}\}$.
Also, design a finite automaton for the same and verify correctness.

— Given language:

$L = \{w \in \{a, b\}^* \mid w \text{ ends with "abb"}\}$

Regular Expression

1. Any string over $\{a, b\} \rightarrow (a+b)^*$
2. Must end with abb
3. Combine

Finite Automaton Design

States:

- $q_0 \rightarrow \text{Start}$
- $q_1 \rightarrow \text{Seen 'a'}$
- $q_2 \rightarrow \text{Seen 'ab'}$
- $q_3 \rightarrow \text{Seen 'abb' (Final)}$

Transitions:

1. $q_0 \xrightarrow{a} q_1$
2. $q_0 \xrightarrow{b} q_0$
3. $q_1 \xrightarrow{a} q_1$

4. $q_1 \rightarrow b \rightarrow q_2$
5. $q_2 \rightarrow a \rightarrow q_1$
6. $q_2 \rightarrow b \rightarrow q_3$
7. $q_3 \rightarrow a/b \rightarrow q_0$

* Verification :

1. Accepts strings ending with 'abb'
2. Rejects other
3. Example accepted : aabb, babb
4. Example : rejected : aba, abba
5. Final state ensure correctness
6. Tracks last 3 characters
7. Hence Valid.

4. Explain the role of Regular Expression in NLP. Discuss how they are used in text preprocessing tasks such as tokenization, normalization and pattern matching with examples.

1. RE is widely used in Natural Language processing (NLP).
2. Helps in text cleaning and preprocessing
3. used before applying ML models.
4. Fast and efficient pattern matching
5. Works on raw text data.
6. language-independent tool.
7. essential in data preparation

* Application

1. Tokenization

1. Splitting text into words / sentences
2. Example: "Hello world" → ["Hello", "world"]
3. Regex: \w+
4. Remove punctuation
5. Identifying words
6. used in NLP pipelines
7. Improves processing

2. Normalization

1. Standardizing text
2. Convert to lowercase
3. Remove special characters
4. Example: "Hi!" → "hi"
5. Regex remove symbol
6. Helps consistency
7. Improves accuracy

3. Pattern Matching

- i. Find specific patterns
- ii. Emails, phone number
- iii. Example: \d{10} for phone
- iv. Extract useful info
- v. used in data extraction
- vi. support automation
- vii. Widely used in chatbots.